

Supplementary Information for: Measuring emergence through collapse of the effective joint possibility space

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This document contains the Supplementary Methods (standard definitions and simulation parameters of the physical and agent-based reference models, and the baseline predictors used in the intervention race), Supplementary Note 1 (the complete ledger of registered misses and protocol amendments), and Supplementary Tables 1–11 (per-seed and per-condition statistics behind every aggregate reported in the main text, the regime-object audits and the per-system qualification table). Every number in the Supplementary Tables is generated directly from the raw experiment outputs by `make_si_tables.py`; no value is transcribed by hand.

Supplementary Methods

Ant-colony commitment model. N ants make repeated trips, each choosing between two routes A and B . Route pheromones φ_A, φ_B are initialized at 1.0; on each trip every ant chooses A with probability $p = (K + \varphi_A)^\alpha / [(K + \varphi_A)^\alpha + (K + \varphi_B)^\alpha]$ with $K = 5$ and $\alpha = 2$ (the standard nonlinear pheromone choice function), after which pheromones update as $\varphi \leftarrow \varphi(1 - \rho) + qn$ with evaporation $\rho = 0.01$, per-capita deposit $q = Q_{\text{total}}/N$, total deposit budget $Q_{\text{total}} = 0.5$ and n the number of ants that chose the route. Because the deposit budget is shared, the fluctuation scale is $N^{-1/2}$, which is the source of the $\ln N$ commitment-time prediction recorded before the finite-size study. Openness is the binary entropy of the route-choice probability. The breakpoint study uses sizes $\{1, 10, 100\}$ with 900 trips; the finite-size study uses sizes 1–500 with a 1,500-trip horizon, checkpoints every 10 trips, 30 episodes per size and 200 bootstrap resamples for the log-law slope.

Kuramoto model. $N = 200$ oscillators follow $\dot{\theta}_i = \omega_i + Kr \sin(\psi - \theta_i)$ plus Gaussian noise of scale $\sigma = 0.05$, where (r, ψ) is the instantaneous mean field, natural frequencies are drawn from $\mathcal{N}(0, 0.5^2)$, and integration uses Euler–Maruyama with $\Delta t = 0.02$ for 600 steps, measured every 10 steps. The openness object is the normalized entropy of the phase distribution over 10 bins, estimated from an ensemble of 20,000 replicates. The breakpoint contrast uses supercritical $K = 1.5$ against subcritical $K = 0.3$; the coupling study spans $K \in \{0.9, 1.1, 1.5, 2.0, 2.5\}$ with two seeds per coupling (Supplementary Table 8).

Potts model. Ferromagnetic q -state Potts model ($q = 2$ and $q = 10$) on a 48×48 periodic lattice with checkerboard Metropolis dynamics (energy -1 per matching nearest-neighbour pair). Temperature is scanned through multiples $\{1.35, \dots, 0.70\}$ of the exact critical temperature $T_c(q) = 1/\ln(1 + \sqrt{q})$,

with 900 thermalization and 300 sampling sweeps per temperature and six replicas from hot and cold starts. Openness is the normalized entropy of the colour-occupancy distribution; the order parameter is $(q f_{\max} - 1)/(q - 1)$ with $f_{\max}$ the dominant-colour fraction. Hysteresis is measured as the hot/cold-start openness gap.

Vicsek model. Standard Vicsek dynamics in a periodic box at density $\rho = 2.0$: each particle moves at speed 0.03 and adopts the mean heading of all particles within radius 1.0 plus uniform angular noise on $[-\eta/2, \eta/2]$. The noise is annealed from $\eta = 3.0$ down to 0.1 (nine levels, 300 steps per level), with population sizes $N \in \{100, \dots, 3200\}$ and three replicas. Openness is the normalized entropy of the heading distribution over 24 bins; the order parameter is $\varphi = |\langle e^{i\theta} \rangle|$.

Swift–Hohenberg pattern selection. One-dimensional Swift–Hohenberg equation $\partial_t u = ru - (1 + \partial_x^2)^2 u - u^3$ plus additive noise of amplitude 10^{-3} , integrated pseudospectrally (linear operator implicit, cubic term explicit) on $M = 256$ modes over a domain of length 64π with $\Delta t = 0.5$ for 500 steps and 20 replicas. The pattern-forming case $r = 0.2$ is contrasted with the stable case $r = -0.2$. Openness is the normalized spectral entropy of the Fourier power distribution; commitment concentrates power in the critical band $|k - 1| \leq 0.1$.

Schelling segregation. Two agent types occupy 45% of a 50×50 periodic lattice each, with 10% vacancies. An agent is unhappy when its same-type fraction among occupied Moore neighbours falls below the tolerance $\tau = 0.35$; unhappy agents move to random vacancies. Dynamics run for 300 sweeps (20 replicas, states saved every 5 sweeps), with a $\tau = 0$ control in which no agent is ever unhappy. Openness is the normalized entropy of the same-type neighbour-count distribution.

Baseline predictors in the intervention race. The predictor race of Fig. 5b compares, for identical intervention episodes, the switch-prediction AUC of five per-episode quantities: the side-openness of the joint policy (the possibility-space object), the object’s position magnitude $|x|$, its velocity magnitude $|v|$, the population attachment (grip) level, and the intervention time index τ itself. All predictors are computed from the same stored rollouts; AUCs are reported per seed and pooled.

Supplementary Note 1: registered misses and amendments

Every experiment in this study was preceded by an internal frozen protocol naming its prediction and falsification clause. This note collects, in one place, every registered miss and every amendment, each of which is also reported at its point of use in the main text or Methods. The corresponding protocol files are released verbatim with the code and raw outputs.

1. **Detector amendments.** The effect-size gate and the saturation-truncated analysis window were added after two preregistered control failures (a flat subcritical false positive and a saturation-knee capture) and re-validated on fresh seeds before any headline adjudication.
2. **Retired instrument extension.** A settled-openness envelope intended to certify non-monotone commitment kept false positives below 0.05 in every validation round but never exceeded 0.71

power against its required 0.90, and was closed under its own preregistered stopping rule without being applied to system data. This is why the coordination-ring onset count is reported at 1/8.

3. **Semi-synthetic injection near-miss.** Recovery power for commitments injected into the ring pipeline’s own estimator saturated at 0.88 against a registered threshold of 0.90; the 30-episode evaluation budget caps single-curve power, and the result is recorded as a near-miss.
4. **A2C partial replication.** The strict parity-thinning clause passed in 3/5 advantage actor–critic seeds, failing solely on thinned-subsample detector power; the threshold was not adjusted.
5. **Representation-equivalence bar missed.** The registered $\geq 90\%$ verdict-preservation bar was missed (8/12 cells); per the registered clause the measured equivalence class is reported instead of blanket invariance (Supplementary Table 3).
6. **Initialization-sweep midpoint.** The initialization-scale sweep moved commitment time monotonically in the signalling system as predicted, but the role-system midpoint was non-monotone—a registered miss reported as such.
7. **Barrier threshold calibration.** The convention deviation-cost clause was registered at 0.50 where the payoff structure caps the measurable cost at $0.4 \times$ the within-population payoff for five-symbol permutation codes; the measured value 0.3996 sits at that structural ceiling and the mis-calibrated clause is reported verbatim.
8. **Source-attribution relocation.** One registered audit prediction on the decomposition ladder missed: under strong knob interaction (pairwise copying beneath a parity mechanism) realized structure genuinely relocates to the individual level, which the ladder reports; this is a property of measuring realized structure, not generator labels.
9. **Grid-resolution amendment.** At the original 25-update grid the neural-replication onsets were under-resolved (1/7 and 1/10); a measurement-density amendment to 5-update resolution, justified by the detector’s validated grid-power curve, was registered before the rerun, and both grids are reported.
10. **Training failure.** A preregistered attempt on the unmodified MPE `simple_spread` benchmark did not meet its frozen competence precondition (conflict-episode coverage 3% after 3,000 PPO updates) and is recorded as a training failure with no theory claim made either way.
11. **Numerical defect.** One float32 entropy clamp that rounded to a degenerate value was found and fixed in code; both affected experiments were rerun with identical seeds and the corrupted outputs archived.
12. **Regime-ensemble agreement bar missed.** The registered verdict-agreement target for plausible alternative regime objects (24/30) was missed at 22/30: the composed speaker–listener channel is a two-stage regime whose late second collapse the monotone detector declines (two cells), one sub-regime cell falls below the evidence threshold, and the across-episode force marginal cancels symmetric per-episode commitments (five cells). No control cell certified an onset (0/20).

13. **Regime-discovery estimator retired and repaired.** The first realization-axis discovery estimator evaluated a kNN on its own training points; the leakage produced spurious collapse, including in untrained controls, and the run is retained as a registered record. A separately registered cross-fitted repair (v2) removed the leakage; the repaired state-trace proxy and a corrected per-episode force candidate still miss the strict onset bar in every seed (best ΔBIC 12.6 against threshold 10 with thinning), so the realization-axis agreement targets (RD2b 0/5 against 4/5; RE1b 0/5 against 4/5) are reported as misses: raw-behaviour proxies under-certify relative to the declared policy-level object.
14. **Untrained-gate onsets.** Untrained grip controls in the discovery audit certify onset because attachment accumulates under random gripping and the structural gate then releases drift commitment for any policy; the registered “untrained $\Rightarrow$ no onset” control prediction was wrong for this system (and only this system), consistent with the paper’s mechanism control attributing the punctuated profile to the gate rather than to learning.

Supplementary Table 1. Frozen-detector operating characteristics on the held-out synthetic benchmark. 200 labelled curves per family per condition; onset power is the detection rate on onset-type positives and the false-positive rate is pooled over the deceleration, gradual and flat control families. The zero power at 12 grid points is the resolution floor cited in the main text.

Operating point	Onset power	False-positive rate
12 points, $\sigma = 0.02$	0.000	0.000
20 points, $\sigma = 0.02$	0.615	0.000
40 points, $\sigma = 0.02$	0.995	0.000
80 points, $\sigma = 0.02$	1.000	0.000
80 points, $\sigma = 0.0$	1.000	0.000
80 points, $\sigma = 0.01$	1.000	0.000
80 points, $\sigma = 0.02$	1.000	0.000
80 points, $\sigma = 0.04$	1.000	0.000
80 points, $\sigma = 0.08$	0.930	0.000

Supplementary Table 2. Adjudication-contract battery. Every headline onset verdict re-adjudicated across 243 analysis contracts per system (saturation fraction $\times$ window $\times$ effect-size gate $\times$ Δ BIC threshold $\times$ grid stride, 3^5 cells). “Onset among gated” is the fraction of adequate-resolution cells whose verdict is unchanged; the location range is the spread of t^* across those cells as a fraction of curve span.

System	Cells	Gated	Onset among gated	t^* range	Range / span
Grip transport (side openness)	243	243	1.00	16–17	1.3%
Ant colony ($N=100$)	243	243	1.00	32–39	7.8%
Learned high-order coordination	243	243	1.00	550–775	11.2%

Supplementary Table 3. Representation-equivalence battery. The openness object recomputed under twelve preregistered measurement representations on byte-identical retrained seeds and adjudicated by the frozen detector. The verdict survives in 8/12 cells; each breaking cell fails for the identifiable reason given in Methods and still locates the commitment window.

Representation	Δ BIC	t^*	Verdict
Convention: population-mean mapping	10.1	250	onset
Convention: per-agent mean entropy	8.4	250	no verdict
Convention: listener-side dual	12.0	250	onset
Convention: behavioural sample (2,048)	10.3	250	onset
Convention: checkpoint-zero reference	10.4	250	onset
Convention: $\varepsilon = 0.01$ truncation	35.6	300	onset
Convention: 5 $\rightarrow$ 3 symbol coarse-graining	7.9	250	no verdict
Grip: policy probabilities	62.2	17	onset
Grip: state quantization 0.1	56.5	17	onset
Grip: state quantization 0.5	111.5	22	no verdict
Grip: realized action counts	42.4	17	onset
Grip: $\varepsilon = 0.01$ truncation	–	–	no verdict

Supplementary Table 4. Regime-ensemble audit. Alternative regime objects enumerated from each environment specification, adjudicated with the frozen detector on byte-identical reruns. Admissible formation-axis alternatives agree with the declared verdict in 22/25 cells; no control cell certifies an onset; the realization-axis force candidates lose the verdict (Methods, ‘Regime-object audits’).

System	Regime object	Class	Onset	Declared	Agree	t^* range
Convention	speaker code (declared)	declared	4/5	4/5	5/5	275–300
Convention	listener code	plausible	4/5	4/5	5/5	250–325
Convention	composed channel	plausible	2/5	4/5	3/5	300–300
Convention	single-meaning sub-regime	plausible	3/5	4/5	4/5	300–425
Convention	pooled symbol marginal	control (erasing)	0/5	4/5	1/5	–
Convention	speaker identity	control (exogenous)	0/5	4/5	1/5	–
Roles	assignment openness (declared)	declared	5/5	5/5	5/5	375–500
Roles	agent 0’s role	plausible	5/5	5/5	5/5	325–500
Roles	owner of role 0	plausible	5/5	5/5	5/5	400–500
Roles	pooled role marginal	control (erasing)	0/5	5/5	0/5	–
Grip	force-sign ensemble marginal	plausible (mis-specified)	0/5	5/5	0/5	–
Grip	$ x $ tertile occupancy	control (erasing)	0/5	5/5	0/5	–
Grip	per-episode force direction (v2)	plausible (registered miss)	0/5	5/5	0/5	–

Supplementary Table 5. Regime-discovery audit. Machine-discovered regime variables (k-means on raw episode records, k by silhouette, one recipe across systems) adjudicated with the frozen detector; controls apply the identical pipeline to the untrained population or policy. Formation-axis disagreements sit at the detector threshold; the grip rows use the cross-fitted v2 estimator; untrained grip controls certify onset because the structural gate delays commitment for any policy (Methods).

System	Seed	k	Declared	t_{decl}^*	Discovered	ΔBIC	t^*	Control
Convention	0	10	onset	300	–	6.9	1375	–
Convention	1	10	onset	275	–	9.8	400	–
Convention	2	10	onset	300	onset	91.8	475	–
Convention	3	10	onset	275	onset	43.9	400	–
Convention	4	10	–	250	onset	23.6	425	–
Roles	0	10	onset	475	onset	64.9	500	–
Roles	1	8	onset	375	onset	59.2	425	–
Roles	2	7	onset	500	–	73.1	550	–
Roles	3	10	onset	400	onset	43.2	375	–
Roles	4	9	onset	500	onset	71.6	525	–
Grip (cross-fitted)	0	2	onset	18	–	7.4	28	onset
Grip (cross-fitted)	1	2	onset	17	–	5.9	28	onset
Grip (cross-fitted)	2	2	onset	16	–	12.6	27	–
Grip (cross-fitted)	3	2	onset	17	–	3.3	28	onset
Grip (cross-fitted)	4	2	onset	17	–	8.3	27	–

Supplementary Table 6. Qualification is three-way, not an entropy drop. Standardized certificate verdicts: amplitude (fraction of reference entropy closed) with the three qualification gates. The single-ant control and the Overcooked occupancy object show that large collapse with a failed gate is not certified as emergence—ordinary individual determinization is excluded by construction.

System	M	Regime-level	Endogenous	Persistent	Verdict
grip realization (16 agents)	0.91	yes	yes	yes	emergent: punctuated
convention formation (10 agents)	1.00	yes	yes	yes	emergent: punctuated
role lock-in (6 agents)	1.00	yes	yes	yes	emergent: punctuated
ant colony N=100	0.89	yes	yes	yes	emergent: punctuated
single ant N=1 (control)	0.92	no	yes	yes	organized, not qualified
TRI-C high-order (3 agents)	0.32	yes	yes	yes	emergent: punctuated
Overcooked occupancy (deep MARL)	0.11	no	yes	yes	organized, not qualified

Supplementary Table 7. Grip-transport flagship, per-seed statistics. Five REINFORCE seeds (primary) and five advantage actor-critic seeds (byte-identical environment). ΔBIC and t^* are from the primary hinge fit; $\dagger$ marks A2C seeds whose strict parity-thinning clause fails on thinned-subsample detector power (the 3/5 partial replication reported in the main text). Final side mean near zero confirms the left/right symmetry of the learned population.

Seed	Success	Plateau (steps)	ΔBIC	t^*	Final side mean
REINFORCE 0	0.999	19	52.7	18	-0.003
REINFORCE 1	0.999	19	45.8	17	0.007
REINFORCE 2	0.995	18	46.2	16	-0.012
REINFORCE 3	0.999	19	52.5	17	-0.013
REINFORCE 4	1.000	18	52.6	17	-0.036
A2C 0	1.000	17	42.2	15	0.029
A2C 1	0.999	18	45.5	16	-0.006
A2C 2 †	1.000	16	42.7	15	0.012
A2C 3 †	0.999	17	44.1	16	0.033
A2C 4	1.000	16	37.7	14	-0.025

Supplementary Table 8. Non-constructed learned systems (tabular policies), per-seed statistics. Ten-agent Lewis signalling population and six-agent division-of-labour task. “Capability cross” is the update at which mutual intelligibility (respectively team success) first reaches 0.9; in every onset seed the breakpoint precedes it.

Seed	Success	ΔBIC	t^*	Capability cross	Verdict
Signalling 0	0.999	23.0	300	925	onset
Signalling 1	0.999	17.8	275	700	onset
Signalling 2	0.999	30.1	300	1025	onset
Signalling 3	0.999	19.7	275	750	onset
Signalling 4	0.999	6.6	250	600	no verdict
Roles 0	1.000	63.0	475	575	onset
Roles 1	1.000	53.6	375	525	onset
Roles 2	1.000	71.2	500	625	onset
Roles 3	1.000	61.4	400	525	onset
Roles 4	1.000	71.7	500	625	onset

Supplementary Table 9. Neural replications, per-seed statistics at 5-update resolution. One-hidden-layer networks replace the tables with environments and recipes fixed (per-agent MLPs for signalling; one shared MLP for all six role agents). Random initialization compresses commitment into the first few hundred updates, which is why these runs are evaluated on the preregistered fine grid. Seeds that never reach the 0.9 capability criterion (final success 0.800) are shown for completeness but excluded from the learned-seed counts in the main text.

Seed	Success	ΔBIC	t^*	Capability cross	Verdict
Neural signalling 0	1.000	50.9	70	255	onset
Neural signalling 1	1.000	82.4	245	400	no verdict
Neural signalling 2	0.800	40.5	80	–	onset
Neural signalling 3	1.000	65.6	120	1290	onset
Neural signalling 4	1.000	35.9	75	295	onset
Neural signalling 5	0.800	21.8	70	–	onset
Neural signalling 6	1.000	114.5	95	240	onset
Neural signalling 7	1.000	25.4	90	280	onset
Neural signalling 8	1.000	96.0	100	295	onset
Neural signalling 9	0.800	26.0	75	–	onset
Neural roles 0	1.000	160.2	180	220	onset
Neural roles 1	1.000	162.3	160	210	onset
Neural roles 2	1.000	83.3	80	130	onset
Neural roles 3	1.000	110.2	135	180	onset
Neural roles 4	1.000	103.2	115	170	onset
Neural roles 5	1.000	95.3	90	140	onset
Neural roles 6	1.000	118.4	120	190	onset
Neural roles 7	1.000	142.5	105	180	onset
Neural roles 8	1.000	113.0	105	155	onset
Neural roles 9	1.000	36.3	45	100	onset

Supplementary Table 10. Overcooked coordination-ring conventions, per-seed. All 8 pooled self-play seeds end committed to one circulation direction (final direction probability at the Laplace floor or ceiling); 1/8 certifies a punctuated onset, the instrument boundary on non-monotone commitment discussed in Methods.

Seed	Final $P(\text{ccw})$	Committed direction
Original 95101	0.97	counterclockwise
Original 95202	0.97	counterclockwise
Original 95303	0.03	clockwise
Extension 95606	0.97	counterclockwise
Extension 95707	0.97	counterclockwise
Extension 95808	0.03	clockwise
Extension 95909	0.97	counterclockwise
Extension 96010	0.97	counterclockwise

Supplementary Table 11. Scaling data. Ant-colony finite-size scaling (median t_{50} crossing and 10–90% width in trips, 30 episodes per size) and the Kuramoto coupling grid (two seeds per coupling K ; t^* in time units). These rows are the points behind Fig. 6a,b.

System / size	t_{50} or t^*	Width	ΔBIC	Verdict
Ant $N = 1$	220	220	25.4	–
Ant $N = 2$	230	270	63.1	–
Ant $N = 5$	280	290	78.7	–
Ant $N = 10$	320	270	16.2	–
Ant $N = 20$	370	300	40.0	onset
Ant $N = 50$	430	280	76.7	onset
Ant $N = 100$	480	280	111.8	onset
Ant $N = 200$	510	290	133.3	onset
Ant $N = 500$	670	290	236.6	onset
Kuramoto $K = 0.9$ (seed 82001)	6.6	–	195.7	onset
Kuramoto $K = 0.9$ (seed 82002)	6.8	–	202.1	onset
Kuramoto $K = 1.1$ (seed 82001)	5.4	–	240.8	onset
Kuramoto $K = 1.1$ (seed 82002)	5.4	–	239.1	onset
Kuramoto $K = 1.5$ (seed 82001)	3.2	–	95.7	onset
Kuramoto $K = 1.5$ (seed 82002)	3.2	–	93.7	onset
Kuramoto $K = 2.0$ (seed 82001)	2.4	–	63.3	onset
Kuramoto $K = 2.0$ (seed 82002)	2.4	–	63.8	onset
Kuramoto $K = 2.5$ (seed 82001)	1.8	–	53.5	onset
Kuramoto $K = 2.5$ (seed 82002)	1.8	–	53.4	onset